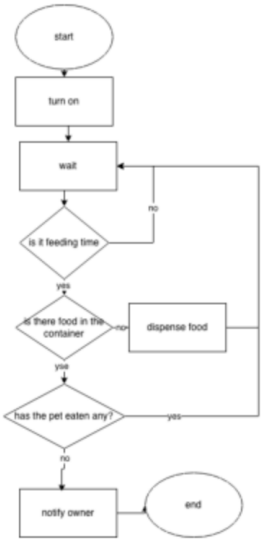
ST1 week 7

Task 1_1

```
<class 'PIL.Image.Image'>
None
RGB
(324, 661)
```



The keras API loads the image for us, it also gives us a bunch if unformation about it before showing us the image

Task 1_2

```
<class 'PIL.Image.Image'>
float32
(661, 324, 3)
<class 'PIL.Image.Image'>
```

This one shows how to type cast it between Pil images and numpy arrays, and shows how they are related and can be read.

Task 1_3

```
<class 'PIL.Image.Image'>
None
RGB
(625, 350)
```

This one shows how we can modify an image and then save the modified image as a separate file, and open that file.

Task 1_4

```
Train (60000, 28, 28) (60000,)
Test ((10000, 28, 28), (10000,))
Train 0 255 33.318421449829934 78.56748998339798
Test 0 255 33.791224489795916 79.17246322228644
```

This one shows us how we can load these two data sets. We can also look at the basic information for it.

Task 1_5

```
Train min=0.000, max=255.000
Test min=0.000, max=255.000
Batches train=938, test=157
Batch shape=(64, 28, 28, 1), min=0.000, max=1.000
```

This one adds another grey scale channel, then normalizes the values from 0-255 to 0-1.

And then it batches it?

Task 1_6

```
Means train=33.318, test=33.791
Data Generator Mean: 33.318
(64, 28, 28, 1) 1.7941157
(60000, 28, 28, 1) -1.9512918e-05
```

Does the same as the other one but centres it

Task 1_7

```
Statistics train=33.318 (78.567), test=33.791 (79.172)
Data Generator mean=33.318, std=78.567
(64, 28, 28, 1) 0.019318787 1.0161676
(60000, 28, 28, 1) -3.4560264e-07 0.9999998
```

this one also turns it into a signle chanell and does other stuff to standardize it